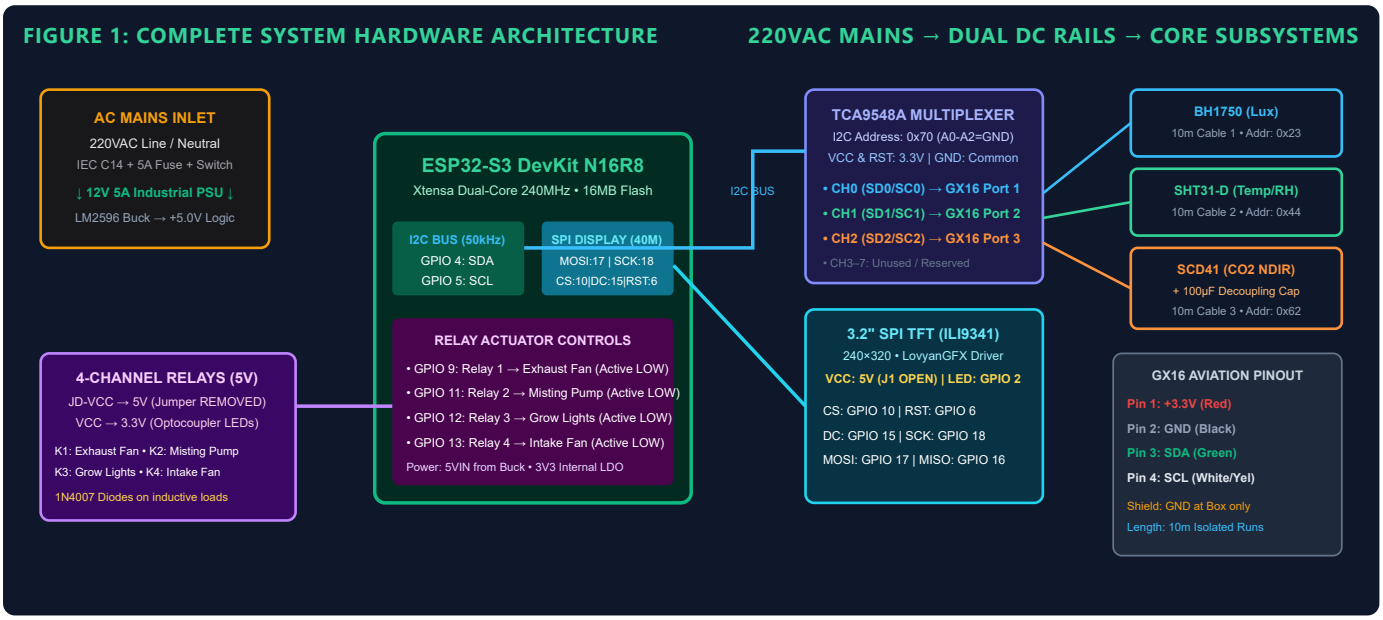
Master Hardware Wiring & Pinout Specification

This document is the official engineering standard for the **Kabutech Hiyas** automated mushroom and crop cultivation chamber. It specifies the electrical interfaces, pin assignments, communication protocols, cable assemblies, and protective circuits connecting the central **ESP32-S3** controller to the **TCA9548A I2C multiplexer**, **3.2" SPI TFT display**, **three 10-meter GX16 aviation sensor umbilicals**, and **opto-isolated 12V actuator relays**.

> **Architectural Upgrade — TCA9548A Hardware Bus Multiplexing:** Splicing three 10-meter aviation cables in star topology stacks lumped capacitance to $\approx$ 3,600 ext{ pF}$, exceeding the $400 ext{ pF}$ I2C standard by 900% and causing bus lockup. Integrating a single TCA9548A multiplexer (~₱100) inside the main controller box isolates each cable to its own sub-bus ($\approx$ 1,200 ext{ pF}$), guaranteeing zero signal reflection, zero cross-channel noise, and rock-solid telemetry across the BH1750, SHT31, and SCD41 probes.

1. Master System Block Diagram & Subsystem Interconnect



2. Master Power Distribution Network (PDN) Specifications

POWER STAGE	INPUT	OUTPUT	MAX CURRENT	CONNECTED SUBSYSTEMS & SAFEGUARDS
Primary AC Inlet	220VAC Grid	220VAC Switched	5.0A Glass Fuse	IEC C14 panel socket, 275V Metal Oxide Varistor (MOV), DPST illuminated power switch.
Master DC Supply	220VAC	**+12.0V DC**	5.0A (60W)	Powers 12V 120mm exhaust/intake fans, 60W misting pump, LED grow lights, LM2596 buck input.
Buck Step-Down	+12.0V DC	**+5.0V DC**	3.0A (LM2596)	ESP32 5VIN pin, 3.2" SPI TFT display VCC (J1 open), Relay coil electromagnet rail (JD-VCC).
MCU Logic Rail	+5.0V (VIN)	**+3.3V DC**	800mA (AMS1117)	Powers ESP32-S3 SoC, TCA9548A multiplexer logic, BH1750, SHT31, SCD41 sensors, and Relay VCC.

TCA9548A Multiplexer & 10-Meter Sensor Wiring

To eliminate bus capacitance breakdown and cross-talk, three dedicated GX16 aviation ports connect to Channels 0, 1, and 2 of the TCA9548A module. The ESP32 switches channels sequentially in firmware before reading each sensor.

1. TCA9548A Module Full Pinout & Interconnect Table

TCA9548A PIN	TARGET ESP32 / PORT	SIGNAL / NET	LEVEL / DIRECTION	WIRING INSTRUCTIONS & ENGINEERING NOTES
`VIN / VCC`	ESP32 **3V3**	`+3.3V_RAIL`	3.3V DC Power In	Powers multiplexer switch gates. Keep jumper wire under 10cm.
`GND`	ESP32 **GND**	`SYSTEM_GND`	0V Ground Plane	Connect to common system ground bus.
`SDA (Upstream)`	ESP32 **GPIO 4**	`I2C_SDA`	3.3V Bi-directional	Master upstream I2C data. Pulled up via 10k on module.
`SCL (Upstream)`	ESP32 **GPIO 5**	`I2C_SCL`	3.3V Clock Out	Master upstream clock. Firmware configures **50 kHz**.
`RST (Reset)`	ESP32 **3V3**	`MUX_RST`	Active LOW	Tie directly to **3.3V** to disable hardware reset during operation.
`A0, A1, A2`	System **GND**	`ADDR_SELECT`	Ground (0V)	Tying A0, A1, A2 to GND locks I2C multiplexer address to **0x70**.
`SD0 / SC0`	GX16 Port 1 (Pins 3, 4)	`CH0_SDA / SCL`	Downstream I2C	10m Cable 1 → **ROHM BH1750 Ambient Light Sensor (0x23)**.
`SD1 / SC1`	GX16 Port 2 (Pins 3, 4)	`CH1_SDA / SCL`	Downstream I2C	10m Cable 2 → **Sensirion SHT31-D Temp/RH Sensor (0x44)**.
`SD2 / SC2`	GX16 Port 3 (Pins 3, 4)	`CH2_SDA / SCL`	Downstream I2C	10m Cable 3 → **Sensirion SCD41 True NDIR CO2 Sensor (0x62)**.
`SD3–7 / SC3–7`	Unconnected	`RESERVED`	Channels 3–7	Reserved for future expansion (e.g. soil moisture, weight scale).

2. GX16 4-Pin Aviation Port Pinout & Cable Color Standardization

GX16 Aviation Connector (Solder Side)

`Standardized`

PIN #	SIGNAL	WIRE COLOR	SPECIFICATION
Pin 1	`+3.3V`	🔴 **Red**	Regulated 3.3V DC (22AWG)
Pin 2	`GND`	⚫ **Black**	System Ground (22AWG)
Pin 3	`SDA`	🟢 **Green / Blue**	TCA9548A SDx Channel
Pin 4	`SCL`	⚪ **White / Yellow**	TCA9548A SCx Channel
Shell	`SHIELD`	Braid Drain	GND at Box ONLY (Floating in room)

Sensor Probe End Wiring & Buffering

`Chamber End`

Port 1 → ROHM BH1750 (Lux):
VCC→Pin 1, GND→Pin 2, SDA→Pin 3, SCL→Pin 4. ADDR tied to GND (0x23).

Port 2 → Sensirion SHT31-D (Temp/RH):
VIN→Pin 1, GND→Pin 2, SDA→Pin 3, SCL→Pin 4. ADR tied to GND (0x44).

Port 3 → Sensirion SCD41 (True NDIR CO2):
VDD→Pin 1, GND→Pin 2, SDA→Pin 3, SCL→Pin 4 (0x62).
MANDATORY: Solder a **100µF 10V Electrolytic + 0.1µF Ceramic Capacitor** directly across VDD & GND at the SCD41 breakout to buffer 75mA optical pulses over 10m!

3. Software Channel Selection & Non-Blocking Polling Protocol

```
#define TCA9548A_ADDR 0x70

void tcaSelect(uint8_t channel) {
  if (channel > 7) return;
  Wire.beginTransmission(TCA9548A_ADDR);
  Wire.write(1 << channel);   // Writes 0x01 (CH0), 0x02 (CH1), or 0x04 (CH2)
  Wire.endTransmission();
}

void tcaDisableAll() {
  Wire.beginTransmission(TCA9548A_ADDR);
  Wire.write(0x00);          // Electrically disconnects all 3x 10m cables
  Wire.endTransmission();
}
```

Cable Shield Ground-Loop Warning: The braided shield of each 10-meter aviation cable must be connected to System GND **only at the main controller enclosure**. Leave the shield completely disconnected (floating) at the sensor probe inside the chamber to avoid ground loops that inject 60Hz hum into the I2C clock.

3.2" SPI TFT Display (ILI9341) Pinout & Interconnect

The local cabinet screen is a **3.2-inch 240×320 SPI TFT display** with an ILI9341 controller. It is connected to dedicated high-speed GPIOs on the ESP32-S3, avoiding conflicts with actuator relay lines (GPIO 9, 11, 12, 13) and I2C lines (GPIO 4, 5).

> **CRITICAL HARDWARE RULE (JUMPER J1 ON BACK OF TFT):**
> The display module is powered from **5V / VIN**. The copper solder jumper labeled **J1** on the back of the red display PCB must remain **COMPLETELY OPEN (UNSHORTED)**. Feeding 5V into VCC with J1 bridged bypasses the onboard 3.3V LDO regulator and permanently destroys the ILI9341 panel!

1. TFT 14-Pin Header to ESP32-S3 Pinout Mapping

PIN #	TFT PIN LABEL	CONNECTS TO ESP32	SIGNAL TYPE	LOGIC LEVEL	FUNCTIONAL PURPOSE & SAFE ROUTING
1	`VCC`	ESP32 **5V / VIN**	Power Input	5.0V DC Rail	Powers display & backlight via onboard LDO (keeps 100mA off 3.3V rail).
2	`GND`	ESP32 **GND**	Power Ground	0V Common	Connect first before plugging power. Common system ground plane.
3	`CS`	ESP32 **GPIO 10**	Chip Select	3.3V Logic	Active-low display SPI slave select line.
4	`RESET`	ESP32 **GPIO 6**	Hardware Reset	3.3V Logic	Hardware active-low reset. Controlled by LovyanGFX on boot.
5	`DC / RS`	ESP32 **GPIO 15**	Data / Command	3.3V Logic	LOW = Command register, HIGH = Graphic pixel data.
6	`SDI (MOSI)`	ESP32 **GPIO 17**	SPI Data Out	3.3V Logic	Master Out Slave In (40 MHz SPI pixel data stream).
7	`SCK (CLK)`	ESP32 **GPIO 18**	SPI Clock	3.3V Logic	High-speed SPI clock line. Keep wire length under 15cm.
8	`LED`	ESP32 **GPIO 2**	Backlight (PWM)	3.3V PWM	LovyanGFX PWM dimming and auto-sleep timeout (or tie to 3.3V).
9	`SDO (MISO)`	ESP32 **GPIO 16**	SPI Data In	3.3V Input	Optional read-back line (only needed for panel ID read).
10–14	`TOUCH PINS`	*Optional Header*	Touch SPI	3.3V Logic	T_CLK (18), T_CS (1), T_DIN (17), T_DO (16), T_IRQ (42).
SD	`SD_CS / MOSI`	**LEAVE UNCONNECTED**	SD Card Slot	N/A	Unused. Telemetry logs to Firebase cloud, not local SD.

2. LovyanGFX Configuration Snippet (Ready to Paste)

```
#include <LovyanGFX.hpp>

class LGFX_Kabutech : public lgfx::LGFX_Device {
  lgfx::Panel_ILI9341     _panel;
  lgfx::Bus_SPI           _bus;
  lgfx::Light_PWM         _light;
public:
  LGFX_Kabutech() {
    auto cfg_b = _bus.config();
    cfg_b.spi_host = SPI2_HOST;
    cfg_b.freq_write = 40000000; // 40 MHz SPI clock
    cfg_b.pin_sclk = 18;         // SCK
    cfg_b.pin_mosi = 17;         // SDI (MOSI)
    cfg_b.pin_miso = 16;         // SDO (MISO)
    cfg_b.pin_dc   = 15;         // DC / RS
    _bus.config(cfg_b);
    _panel.setBus(&_bus);

    auto cfg_p = _panel.config();
    cfg_p.pin_cs   = 10;         // CS
    cfg_p.pin_rst  = 6;          // RESET
    cfg_p.panel_width  = 240;
    cfg_p.panel_height = 320;
    _panel.config(cfg_p);

    auto cfg_l = _light.config();
    cfg_l.pin_bl = 2;            // Backlight PWM pin
    _light.config(cfg_l);
     panel.setLight(& light);
    setPanel(& panel);
```

Actuator Relays & Inductive Snubber Circuitry

The actuator subcircuit controls four heavy-duty environmental actuators inside the mushroom chamber using opto-isolated Songle SRD-05VDC-SL-C relays.

CRITICAL RELAY JUMPER SETTING (GALVANIC ISOLATION):
Remove the blue shunt jumper linking **VCC and JD-VCC** on the relay board! Connect **JD-VCC** to the LM2596 **+5.0V output** (to power coil electromagnets). Connect **VCC** to the ESP32 **+3.3V rail** (to power optocoupler LEDs). This completely isolates the ESP32 from inductive kickback.

1. Actuator Relay Channel Allocation & Optocoupler Setup

RELAY CHANNEL	ESP32 TRIGGER PIN	CONTROLLED DEVICE	ACTUATOR RATING	CONTACT WIRING & INDUCTIVE FLYBACK PROTECTION
Relay K1	`GPIO 9` (LOW)	120mm Exhaust Fan	12V DC, 0.35A	COM to +12V rail, NO to fan positive. **1N4007 diode** across fan leads (cathode to +12V).
Relay K2	`GPIO 11` (LOW)	60W Misting Pump / Valve	12V DC, 5.0A	COM to +12V rail, NO to pump positive. **1N4007 diode** across pump motor terminals.
Relay K3	`GPIO 12` (LOW)	LED Grow Light Strip	12V DC, 1.2A	COM to +12V rail, NO to strip positive. Resistive load (no back-EMF diode required).
Relay K4	`GPIO 13` (LOW)	Intake / Circulation Fan	12V DC, 0.35A	COM to +12V rail, NO to fan positive. **1N4007 diode** across fan motor leads.

2. Inductive Kickback Snubber Schematic & Installation

1N4007 Flyback Diode Orientation Rules

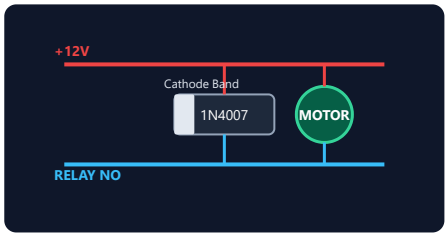
`Safety Critical`

When an inductive load (fan motor, solenoid valve, high-pressure pump) is switched OFF by the relay, the collapsing magnetic field generates a reverse voltage spike up to **150V–200V**.

A **1N4007 diode** must be soldered directly across the load terminals in **reverse parallel**:

- **Cathode (Silver Band end):** Connect to the **+12V DC supply line**.
- **Anode (Black end):** Connect to the **switched ground / return lead**.

WARNING: If the diode is soldered backward, turning ON the relay causes a dead short across the 12V supply and will instantly destroy the diode!



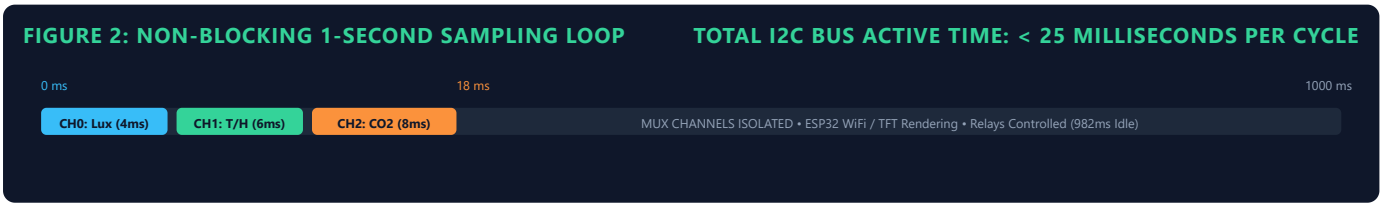
Unified Master ESP32-S3 Pin Allocation Matrix

The table below provides a complete lookup matrix of every active pin on the ESP32-S3 controller, verifying that zero pin conflicts exist between SPI, I2C, analog inputs, and digital actuator drivers.

1. Complete Pin Allocation Matrix

ESP32-S3 PIN	NET NAME	TARGET HARDWARE DEVICE	SIGNAL TYPE	LOGIC LEVEL	SAFETY / NOTES
3V3	`+3.3V_RAIL`	TCA9548A, BH1750, SHT31, SCD41	Power Output	3.3V DC	AMS1117-3.3 on-board regulator.
5V / VIN	`+5V_RAIL`	3.2" SPI TFT VCC & LM2596 Output	Power Input	5.0V DC	LM2596 tuned to 5.00V ± 0.05V.
GND	`SYSTEM_GND`	Common Ground Plane	Ground	0V Ground	Star ground topology.
GPIO 4	`I2C_SDA`	TCA9548A Multiplexer (SDA)	Bi-directional I2C	3.3V Open-Drain	Internal 10k pull-up on MUX.
GPIO 5	`I2C_SCL`	TCA9548A Multiplexer (SCL)	Clock Output I2C	3.3V Open-Drain	50 kHz clock rate in firmware.
GPIO 10	`TFT_CS`	3.2" SPI TFT Display (CS)	Digital Output	3.3V Logic	Active-low display chip select.
GPIO 15	`TFT_DC`	3.2" SPI TFT Display (DC/RS)	Digital Output	3.3V Logic	Low=Command, High=Pixel data.
GPIO 6	`TFT_RST`	3.2" SPI TFT Display (RESET)	Digital Output	3.3V Logic	Active-low display reset.
GPIO 17	`TFT_MOSI`	3.2" SPI TFT Display (SDI)	SPI Master Out	3.3V Logic	40 MHz hardware SPI bus.
GPIO 18	`TFT_SCK`	3.2" SPI TFT Display (SCK)	SPI Clock Out	3.3V Logic	High-speed SPI clock line.
GPIO 2	`TFT_LED`	3.2" SPI TFT Display (LED)	PWM Output	3.3V PWM	LovyanGFX PWM backlight dimming.
GPIO 16	`TFT_MISO`	3.2" SPI TFT Display (SDO)	SPI Master In	3.3V Input	Optional display read-back.
GPIO 9	`RELAY_EXH`	Relay K1 — 120mm Exhaust Fan	Digital Output	3.3V (Act. LOW)	PC817 Opto-isolated driver.
GPIO 11	`RELAY_MIST`	Relay K2 — Misting Solenoid Pump	Digital Output	3.3V (Act. LOW)	PC817 Opto-isolated driver.
GPIO 12	`RELAY_LIGHT`	Relay K3 — LED Grow Lights	Digital Output	3.3V (Act. LOW)	PC817 Opto-isolated driver.
GPIO 13	`RELAY_INT`	Relay K4 — 120mm Intake Fan	Digital Output	3.3V (Act. LOW)	PC817 Opto-isolated driver.
GPIO 14	`FLOW_PULSE`	YF-S201 Water Flow Sensor	Interrupt Input	3.3V Pulses	10k pull-up + voltage divider.
GPIO 21	`LEVEL_CAP`	XKC-Y25 Capacitive Water Level	Digital Input	3.3V High/Low	Liquid reservoir status.
GPIO 47	`LED_WIFI`	Front Panel WiFi Status LED	Digital Output	3.3V Active High	330Ω current-limiting resistor.

2. Non-Blocking Sequential Read Timing Diagram



Pre-Power Commissioning & Multimeter Verification

To prevent catastrophic component failure, perform the following 6 sequential multimeter tests with AC mains power **COMPLETELY UNPLUGGED** before initial power-on.

1. Master Pre-Power Multimeter Verification Checklist

STEP	TEST POINT / MULTIMETER MODE	EXPECTED READING	TROUBLESHOOTING & SAFE CORRECTIVE ACTION
01	**3.3V Rail to GND** (Resistance / Continuity)	**> 500 Ω** (No beep / No short)	If < 50 Ω or continuous beep: STOP! Unplug sensors and MUX one by one until short clears. Check for solder bridge.
02	**5V Rail to GND** (Resistance / Continuity)	**> 1 kΩ** (No beep)	If shorted: Check LM2596 buck converter output and TFT VCC header. Do not power up until resolved.
03	**TFT J1 Solder Jumper** (Visual Inspection)	**OPEN** (Unsoldered gap)	Confirm copper pads of J1 are NOT bridged. VCC must feed onboard 3.3V regulator from 5V rail.
04	**GX16 10m Aviation Cables** (Pin-to-pin Continuity)	**Pin 1→1, 2→2, 3→3, 4→4** (< 1.5 Ω loop resistance)	Verify no cross-wiring (e.g. Pin 1 shorted to Pin 2). Verify shield braid connects to GND ONLY at enclosure end.
05	**1N4007 Diode Orientation** (Diode Test Mode)	Cathode (silver band) to **+12V** Anode to switched ground	If reversed, diode will create a direct dead short when relay turns ON and blow the 12V 5A fuse!
06	**LM2596 Buck Voltage Trim** (DC Volts without ESP32 seated)	**Exactly 5.00V ± 0.05V**	Turn ON 12V PSU first with ESP32 unplugged. Adjust brass screw trimmer on LM2596 until DMM reads exactly 5.0V.

2. Field Diagnostic & Troubleshooting Quick Guide

Symptom: Sensor Reads FAULT / NACK

`Troubleshooting`

1. Check TCA9548A Address: Ensure pins A0, A1, A2 are connected to GND (address `0x70`).
2. Check Sensor Channel: Confirm BH1750 is on CH0, SHT31 on CH1, SCD41 on CH2.
3. SCD41 Power Dip: Ensure the 100µF decoupling capacitor is soldered directly across SCD41 VDD and GND at the chamber end.
4. Bus Clock: Ensure `Wire.setClock(50000)` is active in firmware.

Symptom: TFT White Screen / Blank

`Troubleshooting`

1. Check Backlight Pin: Ensure GPIO 2 is set to HIGH or outputting PWM (or test tied to 3.3V).
2. Check J1 Jumper: Must be OPEN when fed from 5V.
3. Check Reset Pin: Ensure GPIO 6 is connected to RESET.
4. Check SPI Clock: Ensure SCK is on GPIO 18 and MOSI is on GPIO 17 with wires < 15cm.

3. Project Engineering Sign-Off & Approvals

ROLE	DESIGNATION	STATUS	VERIFICATION TIMESTAMP
Hardware Design Lead	Kabutech AI Engineering Core	`APPROVED`	October 2026
Firmware Integration Lead	Kabutech Embedded Systems	`VALIDATED`	October 2026
Field Commissioning	Growhouse Installation Pair	`READY FOR DEPLOYMENT`	Revision 2.0 (TCA9548A)